8 Elements and chemical bonding

1.Stable electronic configurations and valency

2012-6.

a. The Table below shows electron configurations of elements **R**, **S**, **T**, **U** and **V**.

Element	Electron Configuration
R	2, 7
S	2, 8, 6
T	2, 8, 2
U	2, 4
V	2

(i) Which elements in the Table belong to period 2 of the periodic table? (2marks)

(ii) Give a reason for the answer in 6.a.(i). (1mark)

2011-4.

a. (i) State three ways in which atoms attain stability. (3 marks)

b. **Table 1** shows atomic numbers and boiling points of some elements represented by letters **D**, **Q**, **T**, **X** and **Z**.

Table 1

Element	Atomic Number	Boiling Point (°C)
D	3	1342
Q	13	2467
T	16	445
X	18	-186
Z	19	760

(i) Identify any two letters that represent elements which belong to period 3 in the periodic table. (2 marks)

(ii) Which element is in gaseous state at room temperature (25°C)? (1 mark)

(iii) What type of bonding would exist when element **Q** reacts with element **T**'? (1 mark)

(iv) Write down the chemical equation for the reaction that would occur between **D** and **T**. (3 marks)

2010-1.

e. Explain why helium, which has 2 valance electrons, is taken as a group 8 element.

2009-1.

a.Define "electron configuration? (1mark)

b. **Figure 1** is a graph of atomic radius across the periods against atomic number for some elements in the periodic table.

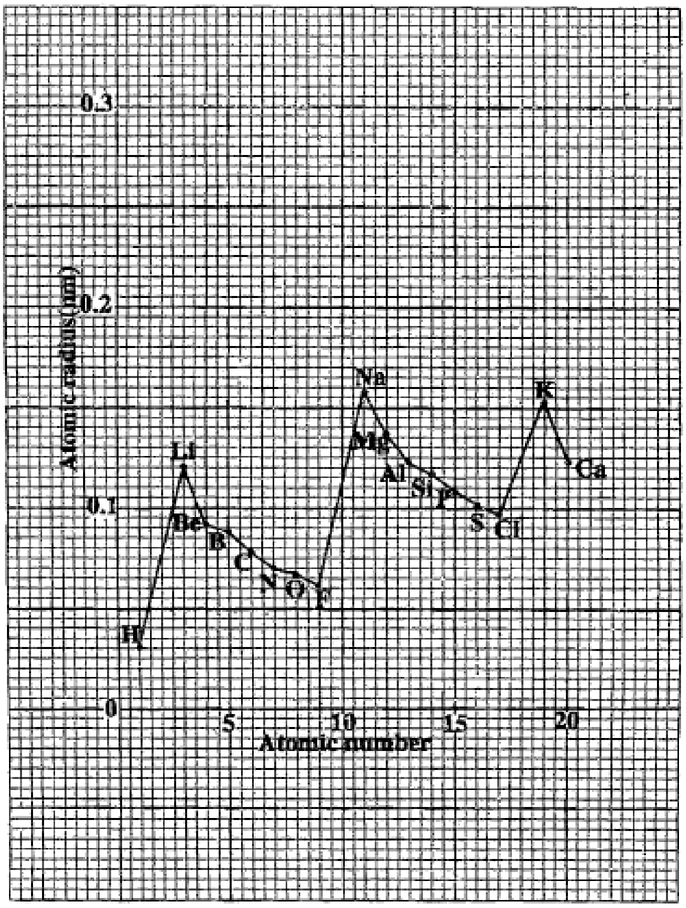


Figure 1

(i) To which group does element O belong? (1mark)

(ii) Give a reason for the answer to 1.b.(i). (1mark)

(iii) Why is there a sudden increase in atomic radius from F to Na? (2marks)

(iv) In terms of atomic radius, explain the difference in reactivity between F and CI. (4marks)

(v) Give two differences between the type of bonding in Lithium metal (Li) and chlorine gas (Cl_2) (2marks)

2009-1.

d. **Table 1** shows the number of valence elections and valences of some elements.

Element	Number of valence electrons	Valency
Li	1	1
Be	2	2
N	5	3
O	6	2

(i) How can element N attain a stable configuration? (1mark)

(ii) Give reason for the answer to 1.d.(i). (1mark)

(iii) What is the formula of a compound that is formed between Li and O? (1mark)

(iv) Give the charge on a Be ion. (1mark)

2008-1.

a. Element **X** has a mass of 89 amu and atomic number 19.

(i) How many protons are in the atom? (1 mark)

(ii) What would happen if element **X** was mixed with water? (1 mark)

(iii) Give a reason for the answer to 1.a.(ii). (1 mark)

2007-5.

a. **Table 1** shows atomic numbers and electron configurations of some elements.

Table 1

Element	Atomic number	Electron configuration
A	18	2, 8, 8
B	10	2, 8
C	20	2, 8, 8, 2
D	12	2, 8, 2
E	2	2
F	9	2, 7

(i) Identify an element that comes first in period 2. (1mark)

(ii) Which two elements can form positive ions? (2marks)

(iii) Give a reason for the answer to 5.a.(ii). (2marks)

(iv) Give any three properties of element **A**. (3marks)

2005-2.

Table 1 shows the arrangement of some elements in the Periodic table.

Table 1

H							He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar
K	Ca						

a. Draw the atomic structure of Cl. (3marks)

b. A certain element could be represented as $^{28}_{14}X$

(i) To which group does X belong? Give a reason. (2marks)

(ii) Identify element **X** in Periodic Table.

2005-4.

d. **Figure 4** is a diagram showing nuclei of two atoms.

Figure 4

(i) Explain why these atoms react in the same way. (3marks)

(ii) What are atoms of this type called? (1mark)

(iii) To which period of the periodic table could each belong? (1mark)

(iv) Explain the answer to d(iii). (1mark)

2004-1.

a. **Table 1** shows particles found in the atoms of four elements.

Table 1

ELEMENT	PROTONS	NEUTRONS	ELECTRONS	MASS NUMBER
Hydrogen (H)	1			1
Carbon (C)			6	12
Nitrogen (N)	7	7		
Sodium (Na)		12	11	

(i) Complete the table by filling the missing numbers. (4 marks)

(ii) Which element in the table will easily form anionic compound? (1 mark)

(iii) Give a reason for the answer to 1.a.(ii) . (2marks)

c.

Table 2 shows elements represented by letters **Q**, **R**, **L**, **M**, **X**, **W**, **Y** and **Z** in the same period of the periodic table.

GROUP	I	II	III	IV	V	VI	VII	VIII
Elements	Q	R	L	M	X	W	Y	Z.

(i) Write the formula of a charged atom of **R**. (1 mark)

(ii) Give the letter of the element in the table which belongs to the halogen family. (1mark)

(iii) Give the latter of an element in the table that would not react with another element. (1mark)

(iv) Give a reason for the answer to 1c(iii) (1 mark)

2003-3.

c. Table 1 shows the first 20 elements of the periodic table.

H							He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar
K	Ca						

(i) Write down the atomic number of Si (1mark)

(ii) Work out the electron configuration of K given that its atomic number is 19 (1mark)

(iii) Draw an electron dot and cross diagram of CO_2 (2marks)

(iv) How can aluminum (Al) attain an inert gas configuration? (1mark)

(v) Explain why the melting points of group VII elements increase with increasing atomic number.

2.Types of bonds and their properties

2012-6.

a. The Table below shows electron configurations of elements **R**, **S**, **T**, **U** and **V**.

Element	Electron Configuration
R	2, 7
S	2, 8, 6
T	2, 8, 2
U	2, 4
V	2

(iii) Give a pair of elements that would form an ionic compound when they react. (2mark)

(iv) Draw an electron dot and cross diagram for the compound formed when **S** combines with **U**. (3marks)

2011-4.

a. (ii) Explain how ionic bonding occurs. (3 marks)

2010-1.

c. (i) State any three properties of metals. (3 marks)

d. **Figure 1** is a diagram of atomic nuclei of elements **R** and **Q**.

Figure 1

(i) Write down the electronic configurations of elements **R** and **Q**. (1 mark each)

(ii) To which period and group of the periodic table does element **R** belong? (1 mark each)

(iii) Draw a dot and cross diagram of the component that would be produced when **R** reacts with **Q**.

2008-1.

b. Magnesium and chlorine can be represented as $^{24}_{12}Mg$ and $^{35.5}_{17}$Cl, respectively.

(i) What are the valencies of magnesium and chlorine? (1 mark each)

(ii) What is the molecular formula of the compound formed as a result of magnesium reacting with chlorine? (2 marks)

c.

(i) Draw an electron dot and cross diagram of carbon dioxide ($C0_2$) given that carbon is in group 4 and oxygen in group 6 of the periodic table. (4 marks)

(ii) What type of bonding exists in carbon dioxide? (1 mark)

(iii) Give a reason for the answer to 1.c,(ii). (1 mark)

2005-2.

c.

(i) Write the chemical formula of the compound formed between Al and O. (1mark)

(ii) What type of bound exists between Al and O atoms in the compound formed in C(i)? Give a reason. (2marks)

2004-1.

a.

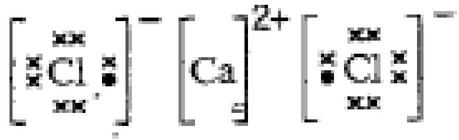
(iv) Work out the molecular mass of methane (CH_4). (2marks)

(v) What kind of chemical bonds are involved in methane? (1 mark)

(vi) Explain the answer to 1.a(v) (3 mark)

b.

The dot and cross diagram of calcium chloride is shown below.



(i) Write the chemical formula of calcium chloride. (1 mark)

(ii) Explain the meaning of the sign 2+ on the Ca atom. (1 mark)

2003-6.

a. Why are metals good conductors of heat? (2marks)

e. State two differences between ionic compounds and covalent compounds. (2marks)

<u>3.Selected elements and their compounds (halogen, nitrogen, sulphur)</u>

2012-6.

b. State any three physical properties of halogens. (3marks)

c. Explain what happened if chlorine is mixed with potassium bromide solution. (2marks)

2010-1.

c. (ii) Explain why potassium is more reactive than sodium (2 marks)

f. State any two uses of sulphates.

2009-1

c.

(i) Mention any two uses of sulphur. (1mark)

(ii) Give any two physical propaties of sulpur. (2marks)

2008-1.

e.

(i) Sulphuric acid (H_2SO_4) can be used as a dehydrating agent. Name the products in the dehydration of sucrose ($C_{12}H_{22}O_{11}$). (1 mark)

(ii) Give any four uses of sulphuric acid. (4 marks)

2006-2.

a. Halogens such as bromine, chlorine and iodine can be prepared by reacting an alkali metal salt with concentrated sulphuric acid in the presence of a catalyst. Name any salt from which each of the following can be prepared. (1mark each)

(i) Br_2 :

(ii) Cl_2 :

(iii) I_2 :

b. State any two properties of halogens. (2marks)

c. Draw an electron shell diagram for a fluorine atom ($^{19}_{9}F$) (2marks)

d. Arrange the elements $^{127}_{53}I$ $^{35.5}_{17}Cl$ and $^{80}_{35}Br$ in order of increasing reactivity. (3marks)

e. Explain the difference in reactivity of the elements in 2.d. (5marks)

f. state any chemical property of sulphur. (1mark)

g. explain, with the aid of diagrams , why rhombic sulpher is more stable than monoclinic sulphur. (4marks)

2005-2.

d. Define the term "allotropes" (1mark)

e. State two allotropes of sulfur. (2marks)

f. Give the halogen used for :

(i) Sterilizing drinking water. (1mark)

(ii) Photography. (1mark)

b. **Table2** shows the atomic numbers, melting points, boiling points and atomic radius of some halogens.

Name of element	Atomic number	Melting point (°C)	Boiling point (°C)	Atomic radius(nm)
Fluorine	9	-220	-188	0.071
Chlorine	17	-101	-34	0.099
Bromine	35	-7	59	0.114
Iodine	53	114	184	0.133

(i) Which element is a liquid at 25°C? (1mark)

(ii) Why does iodine have the biggest radius? (1mark)

(iii) Work out the effective nuclear charge for fluorine. (2marks)

(iv) Mention nay two chemical properties of halogens. (2marks)